

8.1 Classwork: Rational Functions and Asymptotes

1. **Warmup.** Consider the function

$$f(x) = \frac{1}{x}.$$

- (a) Complete the table. Give decimals to 3 significant figures where needed.

x	-1000	-10	-0.1	-0.001	0.001	0.1	10
$f(x)$							

- (b) Describe what happens to $f(x)$ as x approaches 0 from the right (positive side), and as x approaches 0 from the left (negative side).

- (c) Describe what happens to $f(x)$ as x becomes very large (positive), and as x becomes very large negative.

- (d) Sketch the graph of f on the axes below. Write down the equations of the vertical asymptote and the horizontal asymptote of f .

2. For each function below, write down the equation of the vertical asymptote and the equation of the horizontal asymptote.

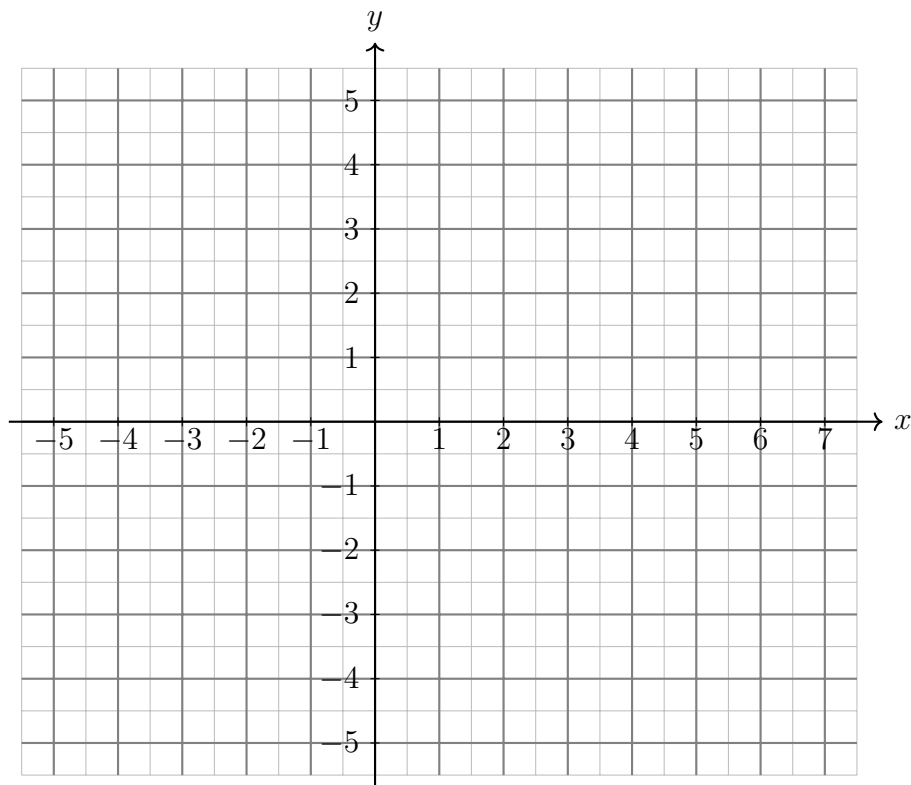
(a) $f(x) = \frac{1}{x-3}$

(b) $f(x) = \frac{1}{x+5} + 2$

(c) $f(x) = \frac{3}{x} - 1$

(d) $f(x) = \frac{2}{x-4} + 1$

3. Sketch the graph of $y = \frac{1}{x-2} + 1$ on the axes below. Plot at least four points on each branch, and show the asymptotes as dashed lines.



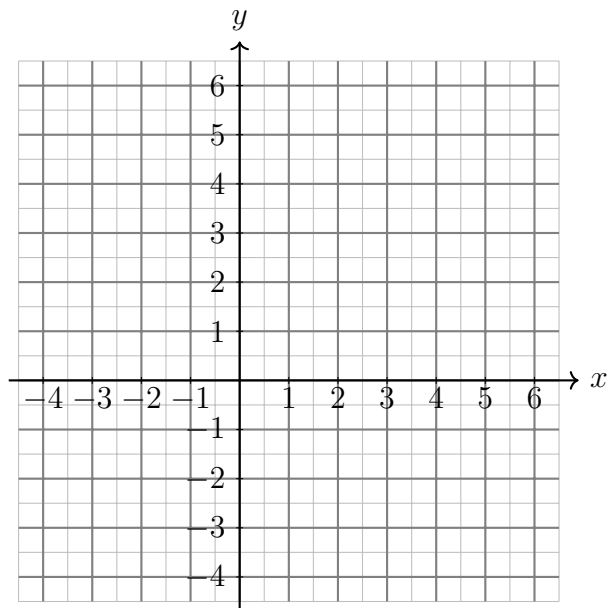
4. Consider the function $g(x) = \frac{1}{x-2} + 1$.

(a) Write down the equations of the vertical and horizontal asymptotes of g .

(b) Find the y -intercept of g .

(c) Find the x -intercept of g .

(d) Sketch the graph of g on the axes below. Label the asymptotes (dashed) and the intercepts.



5. Let $g(x) = \frac{1}{x-2} + 1$, the function from Problem 4.

(a) Find $g^{-1}(x)$. Show each step of your algebra.

(b) Write down the equations of the vertical and horizontal asymptotes of g^{-1} .

(c) Compare the asymptotes of g (from Problem 4(a)) with the asymptotes of g^{-1} .
What do you notice?

(d) State the domain and range of g^{-1} .